

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Resolving Forces Inclined Planes

Slope questions become straight when axes run along and normal to the plane.

RESOLVING FORCES, INCLINED PLANES · M1 · 08

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DR ESLAM AHMED · CAIRO UNIVERSITY FACULTY OF ENGINEERING

The Map

This strategy booklet was mined from the local Mechanics 1 topic problem/answer PDFs, Dr Eslam's source notes, and the WME01 website answer bank. Mechanics questions become reliable when the diagram, sign convention, and equation family are chosen first.

01 AXES: Slope Axes

TRIGGER *inclined plane*

02 NORM: Normal Reaction

TRIGGER *reaction on inclined plane*

03 SLIDE: Sliding Equation

TRIGGER *acceleration on slope*

04 LIMIT: Limiting Friction On Slope

TRIGGER *rough inclined plane, about to move*

BANK EVIDENCE Local pages: 43 problem, 68 answer, 19 notes. Website primary entries: 19. Website linked entries: 29.

01 AXES: Slope Axes

TRIGGER *inclined plane*

BECOMES Resolve parallel and perpendicular to the plane.

FIRST LINE TO WRITE

$mg \sin \theta$ down slope, $mg \cos \theta$ into plane

SIMPLEST STRATEGY

1. Align axes with the plane.
2. Xmark weight components.
3. Equation parallel.
4. Second equation perpendicular.

WORKED MODEL

On a 30° slope, the down-slope component is $mg \sin 30^\circ$.

HARD VARIANTS

1. The normal reaction is perpendicular to the plane.
2. If the angle is given to the vertical, redraw the triangle.
3. Keep signs according to up/down slope choice.

— BOTTOM LINE

Slope axes remove unnecessary diagonal forces.

02 NORM: Normal Reaction

TRIGGER *reaction on inclined plane*

BECOMES Use perpendicular balance because acceleration perpendicular to the plane is zero.

FIRST LINE TO WRITE

$$\sum F_{\perp} = 0$$

SIMPLEST STRATEGY

1. Name perpendicular axis.
2. Oppose into/out of plane components.
3. Reaction is unknown.
4. Make equation.

WORKED MODEL

■ If no other perpendicular force acts, $R = mg \cos \theta$.

HARD VARIANTS

1. Angled pulls or pushes change R .
2. No perpendicular acceleration does not mean no perpendicular forces.
3. Friction depends on this R , so find it first.

— BOTTOM LINE

The normal reaction is found perpendicular to the surface.

03 SLIDE: Sliding Equation

TRIGGER *acceleration on slope*

BECOMES Resolve along the plane and use $\sum F = ma$.

FIRST LINE TO WRITE

$$\sum F_{\parallel} = ma$$

SIMPLEST STRATEGY

1. Select positive direction.
2. List along-plane forces.
3. Insert signs.
4. Do ma .
5. Evaluate acceleration.

WORKED MODEL

Down a smooth slope: $mg \sin \theta = ma$, so $a = g \sin \theta$.

HARD VARIANTS

1. Friction acts opposite motion or impending motion.
2. If motion is up the slope, weight component and friction may both act down slope.
3. Use SUVAT after acceleration is found.

— BOTTOM LINE

The along-plane equation drives the motion.

04 LIMIT: Limiting Friction On Slope

TRIGGER *rough inclined plane, about to move*

BECOMES Set friction to μR only at the limiting case.

FIRST LINE TO WRITE

$$F = \mu R$$

SIMPLEST STRATEGY

1. Locate impending motion.
2. Insert friction opposite it.
3. Make $F = \mu R$.
4. Include weight component.
5. Tidy inequality/equality.

WORKED MODEL

▮ About to slide down: friction acts up the slope.

HARD VARIANTS

1. For rest, friction may be less than μR .
2. If an applied force acts, it may change both R and the limiting friction.
3. Check whether the question asks greatest or least force.

— BOTTOM LINE

Limiting friction is a threshold, not a default.

Quick Recognition Table

WHEN YOU SEE	FIRST LINE TO WRITE
inclined plane	$mg \sin \theta$ down slope, $mg \cos \theta$ into plane
reaction on inclined plane	$\sum F_{\perp} = 0$
acceleration on slope	$\sum F_{\parallel} = ma$
rough inclined plane, about to move	$F = \mu R$

Hard Variant Checklist

STRATEGY	CHECK BEFORE FINAL ANSWER
AXES: Slope Axes	The normal reaction is perpendicular to the plane. If the angle is given to the vertical, redraw the triangle. Keep signs according to up/down slope choice.
NORM: Normal Reaction	Angled pulls or pushes change R . No perpendicular acceleration does not mean no perpendicular forces. Friction depends on this R , so find it first.
SLIDE: Sliding Equation	Friction acts opposite motion or impending motion. If motion is up the slope, weight component and friction may both act down slope. Use SUVAT after acceleration is found.
LIMIT: Limiting Friction On Slope	For rest, friction may be less than μR . If an applied force acts, it may change both R and the limiting friction. Check whether the question asks greatest or least force.